Introduction

A watershed acts as a distinct hydrological system where mass balance principles can be used to track the movement and storage of water. The basic water balance equation assumes that anange in storage (?S) within a catchment is equal to the inputs minus the outputs. In most natural watersheds, the primary input is precipitation while the outputs consist of streamflow discharge, or runoff (Q), and evapotranspira (ET) (Dralle et al., 201

Evaluating this mass balance is critical for understanding how different landscapes respond to storm events and seasonal changes. Factors such as the physical size of the catchment and its subsurface storage capacity directly influence how quickly water enters the stream network, which is visualized using a hydrograph. This report connects the fundamental principles of watershed storage using a small scale paper towel experiment to the hydrograph characteristics and mass balance of two real world catchments in Northern California: Elder Creek and Dry Creek.

Hypothesis

At the experimental scale, it is hypothesized that the larger paper towel will have a higher storage capacity, resulting in a hydrograph with a delayed peak and a smoother, more gradual discharge slope compared to the smaller towel.

At the catchment scale, it is expected that Elder Creek and Dry Creek will exhibit distinctly different hydrograph behaviors and water partitioning (runoff vs. evapotranspiration) due to differences in their landscape storage capacities, despite experiencing similar climatic conditions.

Methods

Part 1, Paper Towel Experiment

Two differently sized paper towels representing a large and small catchment were placed on a tilted surface to allow for runoff collection. A simulated rainstorm was applied using a water bottle, and the weight of the runoff pan was recorded every 15 seconds. The initial catchment mass (76 g for the large towel, 48 g for the small towel) and the total mass of the applied water were recorded. Instantaneous discharge rates were then calculated from the accumulated runoff over time.

Part 2, Real World Catchment Data

Daily precipitation, streamflow (runoff), and potential evapotranspiration (PET) data for Elder Creek and Dry Creek were imported into Python using the Pandas library. The data was sourced from Dralle et al. (2018). Hydrographs were plotted to compare both long term and storm scale runoff responses. An annual water balance was then calculated for the 2016 water year. To solve for actual evapotranspiration (ET), the

assumption was made that the change in interannual storage is zero ($\Delta S = 0$), allowing ET to be computed as the residual of precipitation minus runoff: $ET = P - Q$.

Results

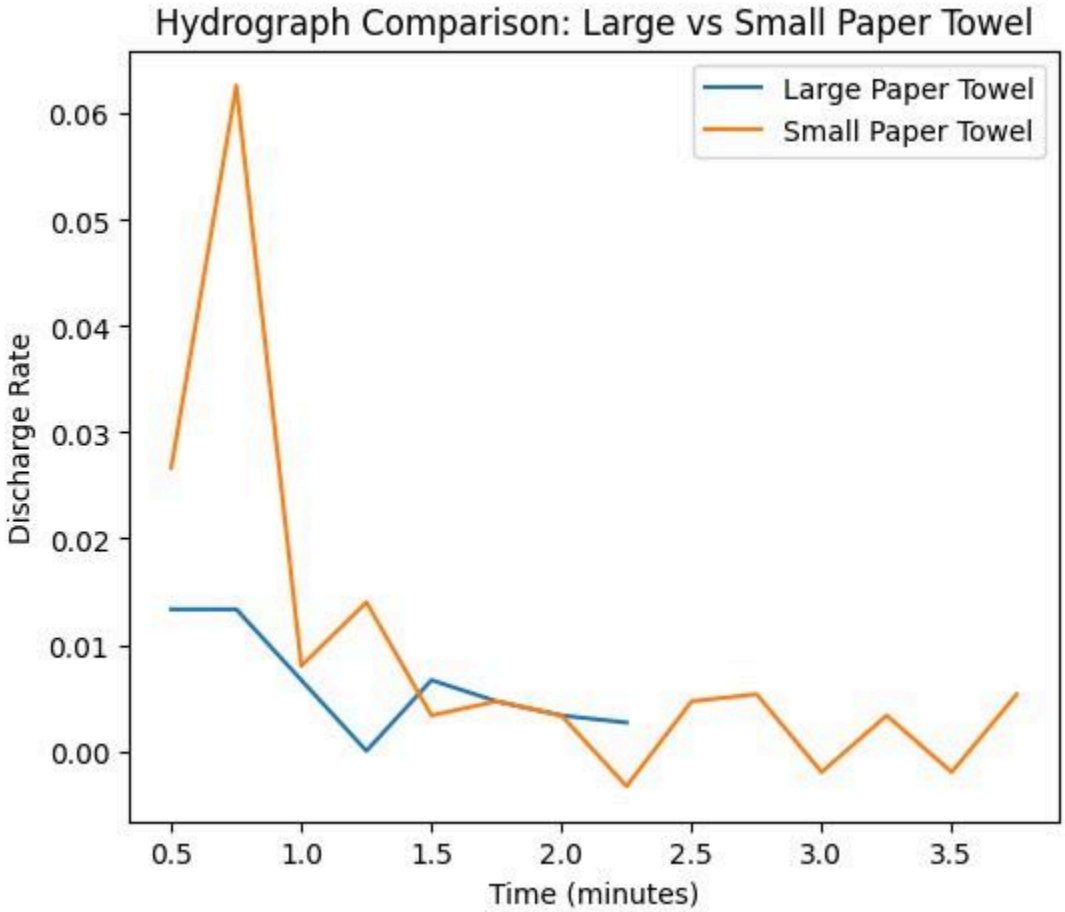

Figure 1. Experimental hydrograph comparing the instantaneous discharge rate (g/min) over time (minutes) for the large and small paper towel catchments. The orange line represents the small paper towel and the blue line represents the large paper towel. The small paper towel shows a sharp, rapid spike in discharge early in the experiment, while the large paper towel maintains a lower, more stable discharge rate throughout, consistent with a higher storage capacity dampening the runoff response.

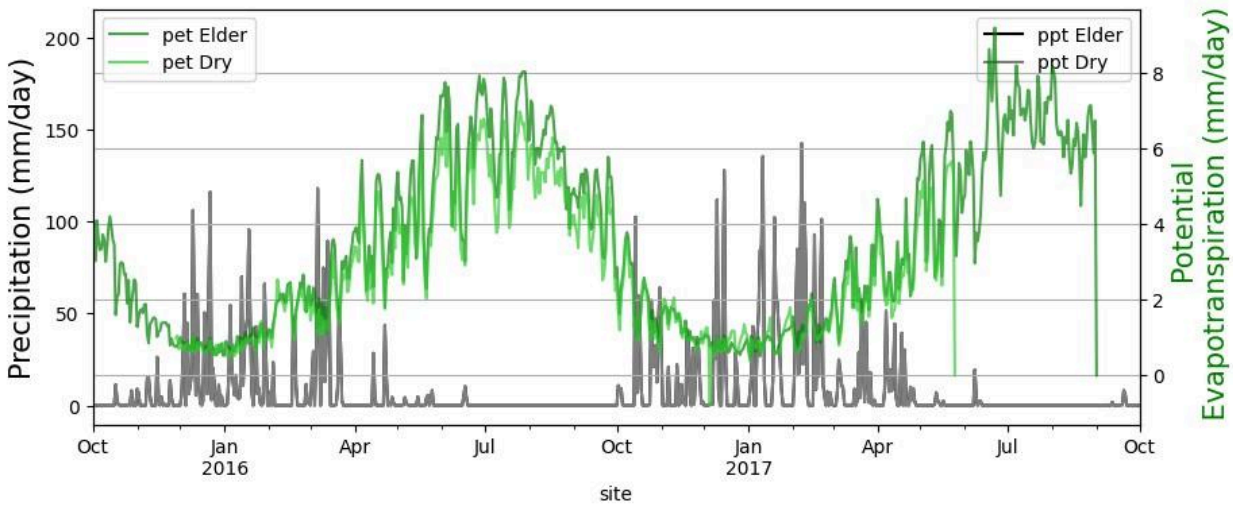

Figure 2. Daily precipitation (mm/day, left y axis, black = Elder Creek, gray = Dry Creek) and potential evapotranspiration (mm/day, right y axis, green = Elder Creek, limegreen = Dry Creek) from October 2015 to October 2017. Precipitation is concentrated in the wet season (October through April), occasionally reaching peaks near 150 mm/day, while PET peaks during the dry summer months (June through September) at roughly 6 to 8 mm/day, when precipitation is essentially absent. This out of phase seasonal pattern is characteristic of a Mediterranean climate.

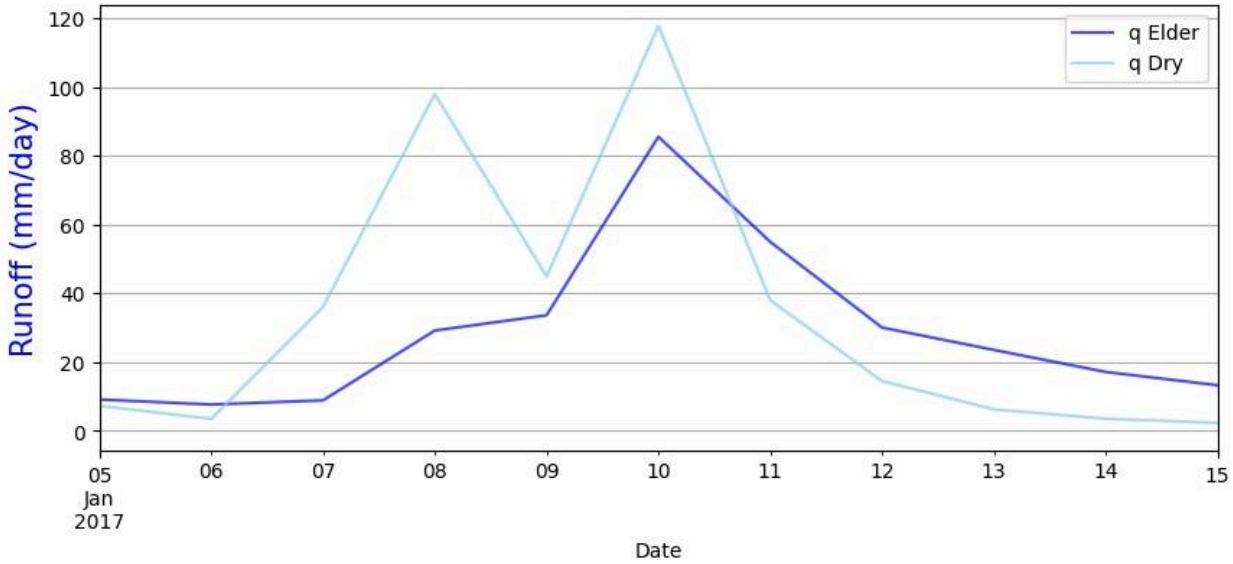

Figure 3. Storm scale hydrograph showing daily runoff (mm/day) for Elder Creek (dark blue) and Dry Creek (light blue) from January 5 to January 15, 2017. Dry Creek exhibits a flashier response, peaking at approximately 120 mm/day on January 10th before steeply declining. Elder Creek responds more gradually, reaching a lower peak of roughly 85 mm/day on the same date, and maintains a higher sustained baseflow through January 12th through 15th as the storm recedes.

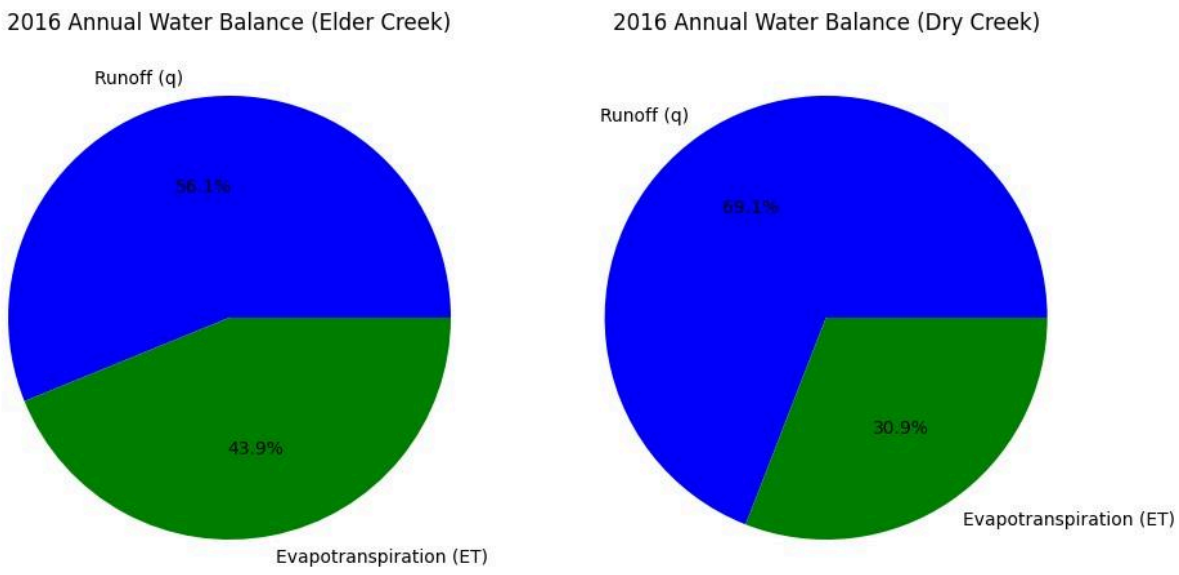

Figure 4. Annual water balance partitioning for the 2016 water year at Elder Creek (left) and Dry Creek (right), expressed as a percentage of total precipitation. Elder Creek partitioned 56.1% of precipitation to evapotranspiration and 43.9% to runoff. Dry Creek partitioned a greater fraction to runoff (69.1%) and a smaller fraction to evapotranspiration (30.9%), indicating more limited water retention capacity.

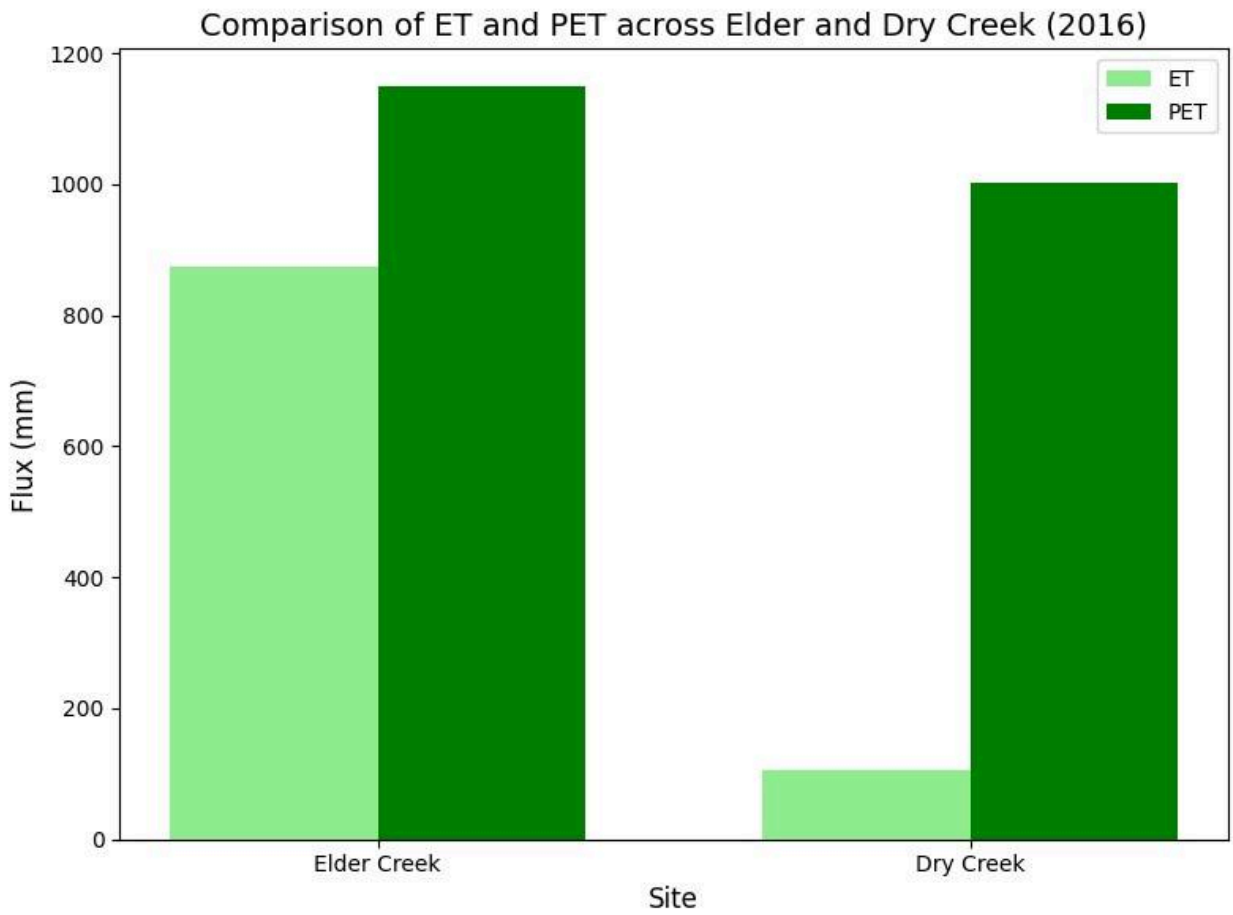

Figure 5. Comparison of calculated actual evapotranspiration (ET, light green bars) and given potential evapotranspiration (PET, dark green bars) in mm for Elder Creek and Dry Creek during the 2016 water year. ET was computed as the residual of the water balance (ET = P − Q). Elder Creek's actual ET of approximately 870 mm approaches its PET of roughly 1,150 mm, suggesting sufficient stored water to sustain vegetation through the dry season. Dry Creek's calculated ET of approximately 100 mm falls far short of its PET of roughly 1,000 mm, indicating severe moisture limitation.

Discussion

The data from both the physical experiment and the real world catchment records support the hypotheses. In the paper towel experiment (Figure 1), the large towel functioned as a catchment with a higher storage capacity. Because it could absorb and retain more water before becoming saturated, it produced a much lower and more stable discharge rate, with the instantaneous discharge rate fluctuating gently between approximately 0.000 and 0.015 g/min throughout the experiment. The small paper towel, with its limited absorptive area, saturated rapidly, forcing nearly all incoming water to immediately convert to surface runoff and producing a sharp discharge spike early in

the experiment. This directly mirrors how storage capacity governs the shape of a hydrograph at the catchment scale.

This physical relationship scales well to the observations at Elder and Dry Creek. As seen in Figure 3, Dry Creek exhibits a flashier storm response, with a rapid, steep peak in runoff that quickly declines after the storm passes, behavior consistent with limited subsurface storage, analogous to the small paper towel. Once its storage capacity is exceeded, water rapidly enters the stream network. Elder Creek, by contrast, likely possesses deeper soils or more permeable bedrock capable of storing larger volumes of water, similar to the large paper towel. This delays the hydrograph peak and sustains baseflow well after the storm event ends (Hahm et al., 2019).

The Mediterranean climate context (Figure 2) adds another layer of importance to storage capacity. Because peak PET occurs in the summer when there is no precipitation, vegetation can only access water that the watershed retained during the winter wet season. As shown in Figure 5, Elder Creek's actual ET comes reasonably close to its PET, suggesting its storage capacity is sufficient to carry moisture through the dry season. Dry Creek's actual ET falls drastically short of its PET, indicating that despite high atmospheric demand, the watershed cannot hold enough water to supply vegetation once precipitation stops. This moisture limited condition is a direct consequence of lower landscape storage capacity.

Conclusions

This lab demonstrated the critical role that storage capacity plays in watershed mass balance. The paper towel experiment confirmed that increased catchment storage buffers storm intensity, limits peak discharge, and produces a smoother hydrograph. The real world data from Elder Creek and Dry Creek reinforced these principles at a larger scale. Catchments with greater storage capacity, like Elder Creek, not only attenuate peak flows during storm events but also retain enough moisture to sustain evapotranspiration throughout the dry summer months. Catchments with lower storage capacity, like Dry Creek, generate flashier runoff responses and experience much drier conditions once seasonal precipitation ceases.

References

Dralle, D., et al. (2018). Storage in catchments.

Hahm, W. J., et al. (2019). Lithologically controlled subsurface critical zone thickness and water storage capacity determine regional plant community composition.